Lab 08 - Capstone - Produce a Verified DevOps Problem Recommendation

Course: DevOps Foundation Training (TGS-2024049184) Version: v4.0 Topic: 08 Estimated time: 35-45 minutes

Workplace scenario

Northstar Retail's checkout service has repeat latency failures after dependency changes. You are the cross-functional problem lead preparing a decision-ready problem record for leadership.

Objective

Apply A1-A6 in one connected case: manage the lifecycle, analyse root cause, prioritise, recommend controls and create monitorable documentation.

What you will produce

A problem_record.md and recommendations.csv that can be audited from detection through verification.

Prerequisites

- R 4.x available as Rscript.
- Work from this lab folder so relative paths resolve.
- Use only the supplied synthetic data; no internet connection or external package is required.
- Preserve the source data. Make experimental changes in a copied file, then restore it.

Folder map

```
Lab-08-capstone-problem-record/
■■■ README.md / README.pdf
■■■ CHECKLIST.md / CHECKLIST.pdf
■■■ capstone_problem_record.R
■■■ data/
■■■ output/            # reproducible evidence produced by the R script
```

Detailed procedure

Step 1 - Review all four capstone datasets and write an initial problem statement that separates impact from assumed cause.

```
Rscript -e 'for(f in list.files("data",full.names=TRUE)) {cat("\n",f,"\n"); print(read.csv(
f))}'
```

Record what you observed before continuing.

Step 2 - Run capstone_problem_record.R to validate referential integrity and generate the first evidence synthesis.

```
Rscript capstone_problem_record.R
```

Record what you observed before continuing.

Step 3 - Open recommendations.csv and refine each action so it names control type, owner, due date, metric and verification threshold.

```
open output/recommendations.csv
```

Record what you observed before continuing.

Step 4 - Open problem_record.md and complete the root-cause and contributing-conditions narrative using only supported evidence.

```
open output/problem_record.md
```

Record what you observed before continuing.

Step 5 - Use the acceptance checklist to test K/A traceability, lifecycle status and recurrence-prevention logic.

```
open CHECKLIST.md
```

Record what you observed before continuing.

Step 6 - Present a three-minute recommendation: problem, evidence, selected controls, trade-off and how success will be verified.

```
open output/problem_record.md
```

Record what you observed before continuing.

Evidence to retain

[ ] output/problem_record.md

[ ] output/recommendations.csv

[ ] completed CHECKLIST.md

[ ] three-minute briefing

Acceptance criteria

- A1-A6 traceability complete
- Cause versus hypothesis distinguished
- At least three proportional controls
- Each action has owner, due date and verification threshold

Verification

All referenced IDs reconcile; the record covers detection through learning; three recommendations include owner, due date and verification; unsupported causal claims are labelled as hypotheses.

Troubleshooting

- cannot open file: confirm your terminal is in this lab folder.
- Rscript: command not found: install R, reopen the terminal and run Rscript --version.
- A validation stop(...) message is evidence that the input broke a stated rule; inspect the named field rather than bypassing the check.
- If an output looks wrong, compare row counts and IDs with the source CSV before changing the analysis.

Clean rerun

Delete only the files inside output/, then run the R script again. The source CSV and script remain unchanged.